Problem Statement #21 — Smart Cities, Transport & Logistics

EV Charging Station Reservation System

1. Requirements Table

1.1 Functional Requirements (FR-001 to FR-005)

ID	Requirement	Priority	Description	Acceptance Criteria
FR-001	Connector & Slot Matching	High	The system shall match the driver's vehicle connector type (CCS, CHAdeMO, Type 2) with available station chargers and reserve a 30-minute charging window.	Pass: Incompatible connector plugs hidden from driver. Fail: Two vehicles booked for same socket concurrently.
FR-002	Real-Time Slot Availability	High	The system shall display real-time availability of charging sockets at nearby stations based on the driver's current location.	Pass: Socket status updates within 5 seconds of a booking. Fail: A booked socket is still shown as available.
FR-003	Dynamic Power Rate Calculation	Medium	The system shall calculate and display the dynamic charging cost based on time of day, station demand, and energy tariff before confirming a booking.	Pass: Quoted price matches the final billed amount within 2%. Fail: Displayed price differs from the rate-card logic.
FR-004	Reservation Confirmation & Notification	Medium	The system shall send a booking confirmation and a reminder notification to the driver before the reserved slot begins.	Pass: Notification received at least 10 minutes before slot start. Fail: No notification sent, or sent after slot start.
FR-005	Station Operator Slot Management	High	The system shall allow station operators to add, remove, or mark chargers under maintenance, updating availability instantly for drivers.	Pass: A charger marked under maintenance disappears from driver search within 5 seconds. Fail: Driver is able to book a charger under maintenance.

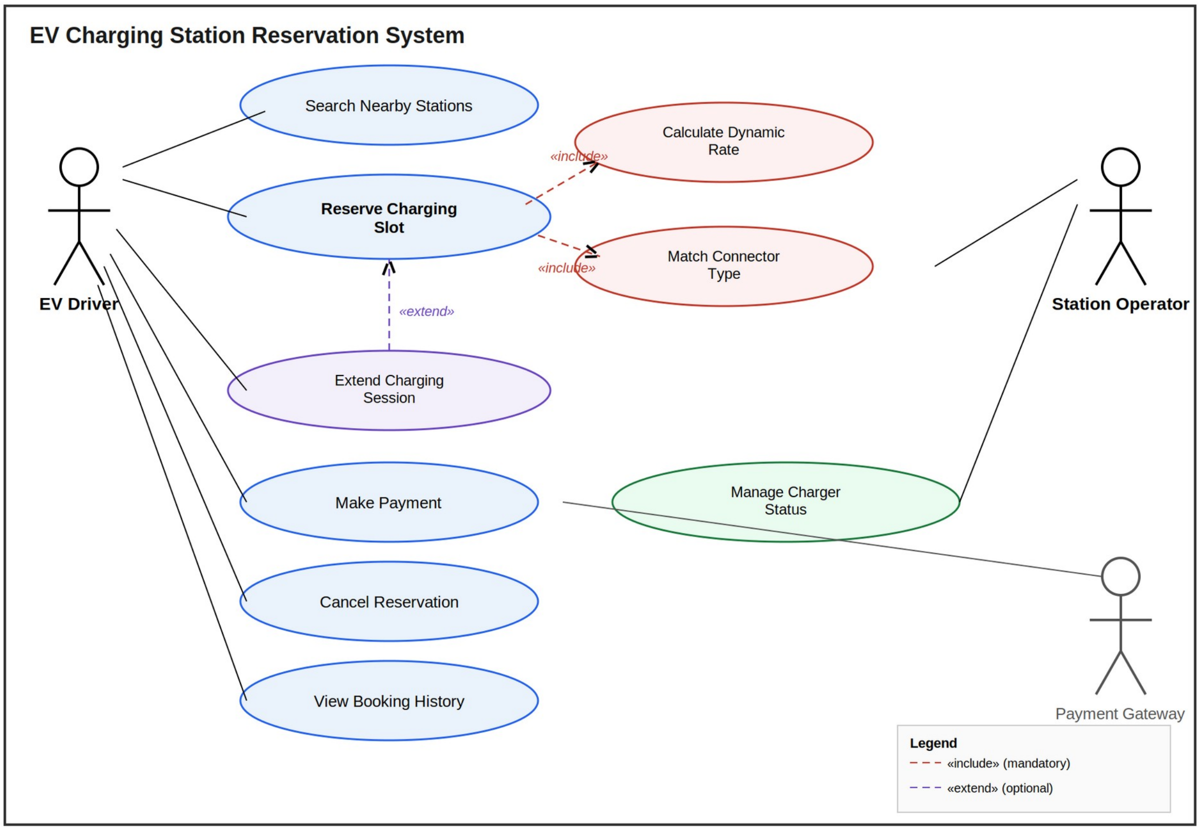
1.2 Non-Functional Requirements (NFR-001, NFR-002)

ID	Type	Description	Priority	Acceptance Criteria
NFR-001	Performance & Security	The system shall automatically release locked sockets and apply a penalty if the driver fails to plug in within 10 minutes of slot start.	High	Pass: Benchmarking tests confirm target latency and security standards under simulated peak load.
NFR-002	Scalability	The system shall support at least 10,000 concurrent booking requests across a city-wide network of charging stations without response time degrading beyond 2 seconds.	Medium	Pass: Load testing confirms sub-2-second response at 10,000 concurrent users. Fail: Response time exceeds threshold or system crashes.

2. UML Use-Case Diagram

Actors: EV Driver (primary), Station Operator (primary), Payment Gateway (secondary/system actor).

Includes at least one «include» relationship (Reserve Charging Slot includes Calculate Dynamic Rate and Match Connector Type — both are mandatory sub-steps of reservation) and one «extend» relationship (Extend Charging Session extends Reserve Charging Slot — optional, only triggered if the driver requests more time and the slot is free).



Note: Recreate this diagram in draw.io or Lucidchart for submission if a native file is required; this rendering follows the same actor/use-case/relationship structure.

3. Use-Case Flow Specification

Use Case: Reserve Charging Slot

Actor: EV Driver

Preconditions:

1. Driver is logged into the app/portal.
2. Driver's vehicle connector type is set in their profile.
3. At least one station with a compatible, available charger exists within the search radius.

Postconditions:

1. A charging slot is locked and reserved for the driver for a defined 30-minute window.
2. The booking confirmation and dynamic rate are recorded in the system.
3. The socket status changes to "Reserved" for all other drivers.

Main Success Scenario:

1. Driver opens the app and searches for nearby charging stations.
2. System displays stations with sockets matching the driver's connector type.
3. Driver selects a station and an available socket.
4. System calculates the dynamic rate for the requested time slot.
5. System displays the rate and estimated charging window to the driver.
6. Driver confirms the reservation.
7. System locks the socket, generates a booking ID, and sends a confirmation notification.
8. Driver arrives and plugs in within the reserved window; charging begins.

Alternate Flow (A1) — Driver Fails to Plug In Within 10 Minutes:

1. At step 8, the driver does not plug in within 10 minutes of slot start.
2. System automatically releases the locked socket (per NFR-001).
3. System applies a no-show penalty to the driver's account.
4. System marks the socket as "Available" again for other drivers.
5. Use case ends.